

Networks Decide for Us

Imputed homophilic networks and the structural premium of diffusion

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Collective behavior spreads on networks

- Collective behaviors — innovations, mobilizations — spread as **contagions** on social networks.
- Real human networks have a specific texture: we cluster with people like us — **multidimensional homophily** [1].
- So a natural question: how much of *where* a behavior ends up is about *who people are*, and how much is about *how they are wired together*?

The gap

- Diffusion ABMs run on *made-up* networks (small-world, scale-free) [2].
- Rich surveys (ATP, GSS) measure *propensities* but carry *no* relational data.
- Threshold models often assume *all* adoption needs peer influence — and when weak-interest groups still tip, the usual explanation is *stochastic noise* that spontaneously instigates a cascade [8].

We ask whether **measured preference and real structure** — not noise, not synthetic graphs — can do that work.

Where we go

Two moves

- **Impute** plausible whole-networks onto real survey respondents from empirical homophily — going from McPherson & Smith's latent “context” term [4] to **full simulable networks**.
- **Measure** the causal premium of that structure on diffusion — and ask *what about the structure* actually matters.

Holding individual propensities *fixed*, does the **shape of the network alone** — not who its members are — decide *how much* collective behavior spreads?

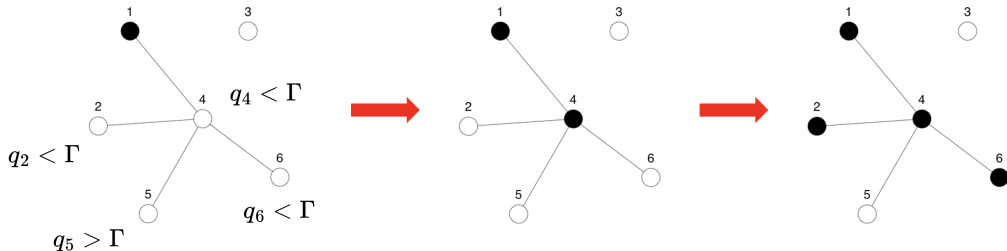
The model

Adoption, part 1 — Rational Choice

- A behavior has an *Intrinsic Utility Level* **IUL**, $\Gamma \in [0, 1]$.
- Each person has a *Minimum Utility Requirement* **MUR**, $q_i \in [0, 1]$, representing their aversion to adopting the behavior.

An agent adopts on its own iff

$$q_i \leq \Gamma \implies a_i = 1$$



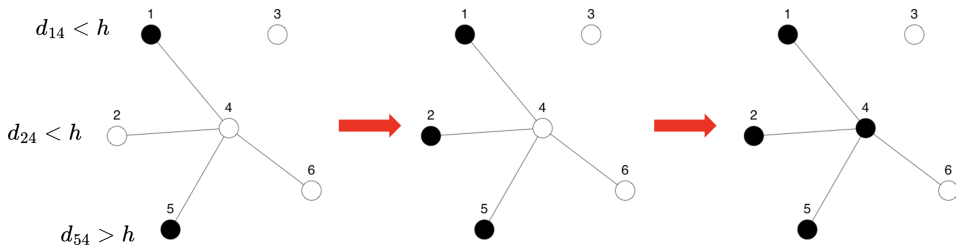
This channel is **topology-blind** — it does not need the network at all.

Adoption, part 2 — Selective Social Influence

Otherwise, an agent adopts by peer contagion if $\tau_i \leq \tilde{E}_i$ [3] — but only peers who are socially *close* [6] count:

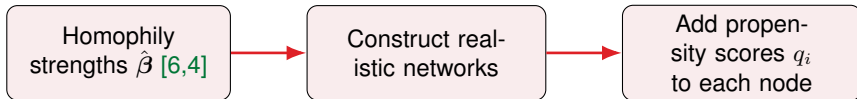
$$\tilde{E}_i \equiv \frac{\sum_{j \neq i} \mathbf{X}_{ij} \tilde{a}_j}{\sum_{j \neq i} \mathbf{X}_{ij}}, \quad \tilde{a}_j = 1 \iff a_j = 1 \wedge U_{ij} \leq \sigma(MSP - d_{ij})$$

- d_{ij} : social (Blau-space) distance; **MSP** (Maximum Social Proximity) sets how far across demographic lines influence can reach.



Imputing structure & running the model

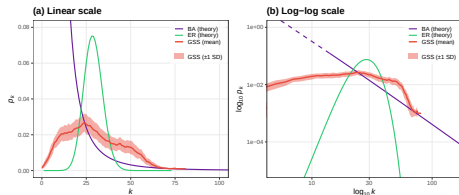
Imputing a network onto a survey



Tie probability is a function of social distance (age, sex, education, race, religion):

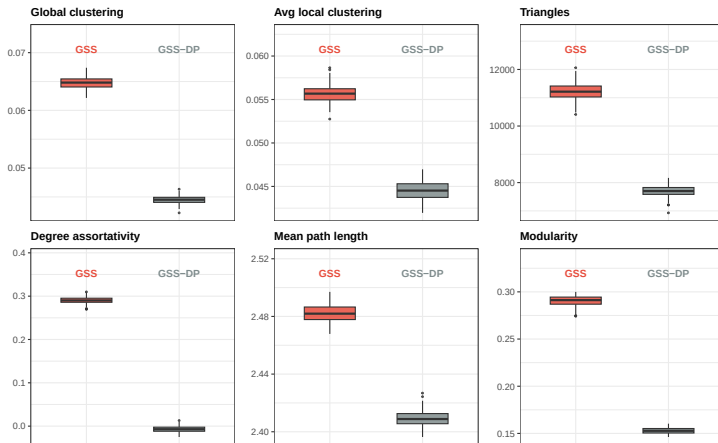
$$\text{logit } \Pr(Y_{ij} = 1) = \alpha + \beta^\top \mathbf{d}_{ij}$$

$\hat{\beta}$ comes from GSS confidant data [6]; propensities q_i come from real survey answers (innovation, collective action).



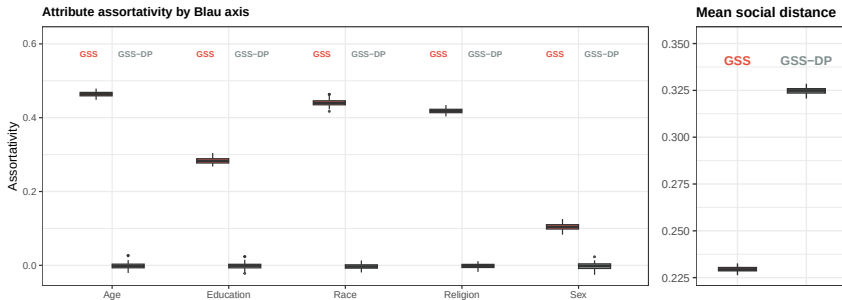
Right-skewed degree structure that Erdős-Rényi / Barabási-Albert models fail to match jointly.

What does “breaking structure” destroy? — topology



100 plausible (GSS) vs. 100 degree-preserving randomized (GSS-DP). Clustering –31%, triangles –31%, modularity –47% — yet path length barely moves, and degree/density/giant-component are identical. The topology degrades, but not uniformly.

What does “breaking structure” destroy? — homophily



Homophily is destroyed: every Blau-axis assortativity collapses to ≈ 0 , and the social distance d_{ij} rises **+42%**.

Comprehensive simulation design

Hybrid social contagion via `netdiffuseR` [7], over plausible **GSS** vs. degree-preserving randomized **GSS-DP** topologies.

- **IUL** (Γ): 41 levels $[0, 1]$. **MSP** (h): 13 levels $[0, 1]$.
- **Thresholds** $\tau_i \sim \mathcal{N}(\mu, \sigma)$: $\mu \in \{.3, .4, .5, .6\}$, $\sigma \in \{.08, .12, .16, .20\}$.
- **Seeding**: random, central, marginal, closeness, eigenvector.

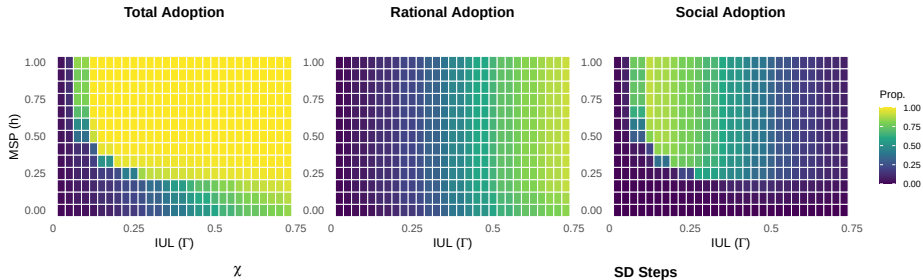
Total: $41 \times 13 \times 4 \times 4 \times 5 \times (\text{runs}) \approx 4 \times 10^6$ **simulations**

We then fit *Big Additive Models* (BAM) to isolate the structural effect:

$$\text{logit}[\mathbb{E}(\Phi)] = \beta_0 + \mathbf{Z}\boldsymbol{\beta} + \beta_{\text{DP}} \mathbb{I}_{\text{DP}} + s(\text{IUL}, \text{MSP})$$

Results

Result 1 — How adoption plays out



Mean adoption over the attractiveness (IUL) – openness (MSP) plane. **Rational** adoption grows smoothly with attractiveness; **social** adoption is more complex, so total adoption stays high over a wide region — *even when the behavior is barely attractive*, social influence alone carries it.

Result 2 — A structural premium

A Big Additive Model isolates the marginal effect of destroying empirical structure:

$$\beta_{\text{DP}} = -0.850^{***} \text{ (OR} = 0.43\text{)}$$

→ randomization cuts adoption odds by
~**57%**

Same people, same propensities — only the **wiring** changed. Homophilic clustering lets social reinforcement build inside local echo chambers.

Total adoption (GSS)	Estimate	OR
(Intercept)	6.534***	—
τ_{mean}	-6.944***	—
τ_{sd}	5.942***	—
<i>Seeding (ref: Random)</i>		
Central	0.120***	1.127
Closeness	0.119***	1.126
Eigenvector	0.118***	1.125
Marginal	-0.067***	0.935
<i>Topology (ref: Plausible)</i>		
Deg.-pres. random	-0.850***	0.427
Deviance explained: 82.2% $N = 4,093,440$		

*** $p < 0.001$. OR = $\exp(\beta)$.

Conclusions

Networks decide for us.

1. **A method:** impute **simulable whole-networks** onto survey respondents from empirical homophily — reusable wherever you have propensities but no ties.
2. **A finding:** holding propensities fixed, empirical structure yields a **+57% adoption premium**.
3. **A mechanism:** what “breaking structure” destroys is, above all, **homophily** — neighbors stop being similar.

Unlike Macy & Evtushenko [8], the idiosyncrasy that drives this is **measured preference**, not stochastic noise: *the same agents, on a different topology, produce a different society.*

References

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Thanks!



Scan: preprint draft (Zenodo)

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Backup — the *shape* of change also differs

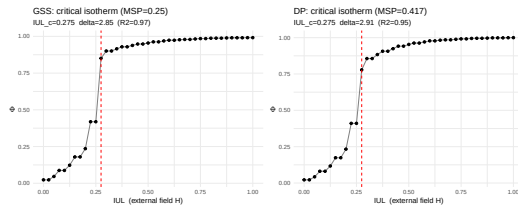
Beyond *how much*, structure governs *how* adoption tips.

$$\chi \sim |MSP - MSP_c|^{-\gamma}$$

$$\Phi \sim |IUL - IUL_c|^{-\delta}$$

Near the tipping point the GSS networks show **mean-field universality**; the randomized ones do not [9].

Critical exponents	GSS	GSS-DP
Suscept. γ	≈ 0.91	≈ 3.9
Isotherm δ	≈ 2.85	≈ 2.91
mean-field	$\gamma=1, \delta=3$	—



Critical isotherms $\Phi(IUL)$ at $MSP=MSP_c$: *both* topologies give $\delta \approx 3$. The field axis does *not* discriminate — only γ (the temperature axis) does.